

Paper A-07: Earth Biodiversity: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

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Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux Φ , constraint C , surface responsiveness S , and structural memory M_s . The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

Notation. In Part IV, C names the active constraint or capacity burden in the system being discussed. Φ is the driving flow, S is the surface response, and M_s is retained structural memory. Λ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

A rainforest and a monoculture cornfield may receive the exact same solar energy flux, yet their thermodynamic stability is vastly different. A minor drought will destroy the cornfield, while the rainforest barely registers the perturbation.

In CDFD, this resilience is not magic; it is pure topological elasticity. Biodiversity is the physical manifestation of an ecosystem's capacity to bend its constraints to absorb incoming flux.

3 Biodiversity as Responsiveness (S)

We govern the resilience of an ecosystem using the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

In an ecological context:

- **Flux (Φ):** The total energetic and material throughput of the environment (sunlight, water, carbon).
- **Constraint (C):** Environmental friction, such as temperature extremes, pathogens, or physical barriers.
- **Responsiveness (S): Biodiversity.** The sheer number of different genetic and phenotypic pathways available to route energy. If one species fails, another immediately fills the niche, ensuring the total throughput Φ is maintained.
- **Memory (M_s):** Ecological specialization. As environments remain stable for millions of years, species over-specialize to maximize efficiency, heavily locking the system into specific inter-dependencies.

3.1 Monocultures and the Mathematics of Fragility

In a highly diverse ecosystem, S is large. If a constraint C spikes (e.g., a new disease), the high S ensures Ψ_s remains stable near 1.

In human-engineered monocultures (e.g., a vast plantation of genetically identical bananas), S is artificially forced to a near-zero minimum. To keep the system producing ($\Psi_s \approx 1$), humans must artificially force the environmental constraints to zero ($C \rightarrow 0$) using massive applications of pesticides and fertilizers.

This state is a thermodynamic house of cards. The moment human intervention ceases and a natural constraint reappears (C spikes), the denominator overwhelms the equation. Because S is fundamentally broken, the system instantly collapses into a $\Psi_s \ll 1$ state of total necrosis.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Conclusion

Biodiversity is not simply a catalog of species; it is the mathematical parameter S that keeps the Earth's biological surface elastically robust against constraint shocks. Using CDFD, we model the hypothesis that replacing complex ecosystems with low-diversity systems guarantees catastrophic topological failure the moment external stabilization forces are removed.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

6 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.

- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdf execution engine, 2026.